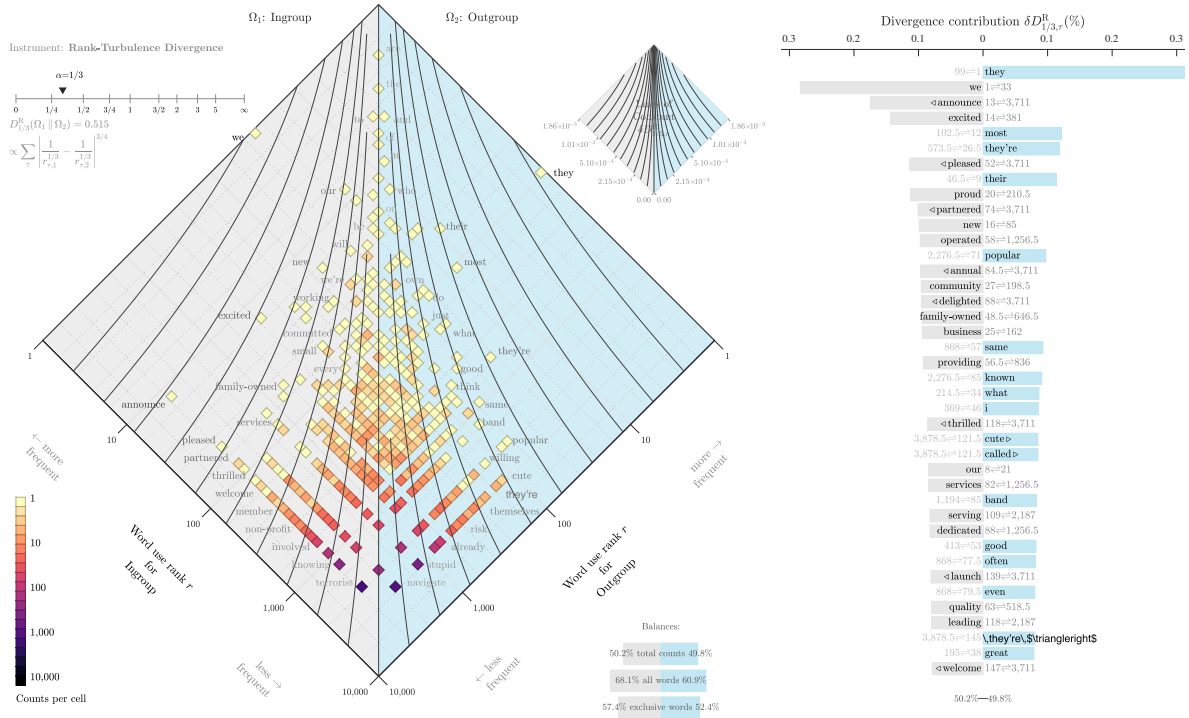




guage in ingroup ("We are") sentences, consistent with ingroup favoritism. In contrast, they generate more negative and lexically diverse language in outgroup ("They are") sentences, reflecting outgroup degradation. Words such as "champions," "believers," "advocates," "reproductive," and "equity" appeared exclusively in ingroup sentences, while "hypocrites," "blame," and "bubble" appeared only in outgroup sentences.

Figure 3 shows the sentiment shifts ingroup/outgroup sentences where ingroup sentences exhibit higher (Φ_{avg}) due to a greater frequency of positive words and fewer negative words compared

to outgroup sentences. Outgroup sentences contain a higher prevalence of negative words ($- \uparrow$) such as "victim", "fear", "obsessed", "problems", "crime", "down", "stuck", "harm", "forget", "blind", "enemies", "vulnerable", "blame", and "resist", all of which decreased Φ_{avg} relative to the baseline (ingroup sentences). This negative shift was further reinforced by a lower frequency of positive words ($+ \downarrow$) in outgroup sentences, such as "we", "free", "proud", "quality", "freedom", and "education". Interestingly, certain negative terms including "fight", "fighting", "criminal", "discrimination", "debt", "critics", and "crime" ($- \downarrow$) ap-

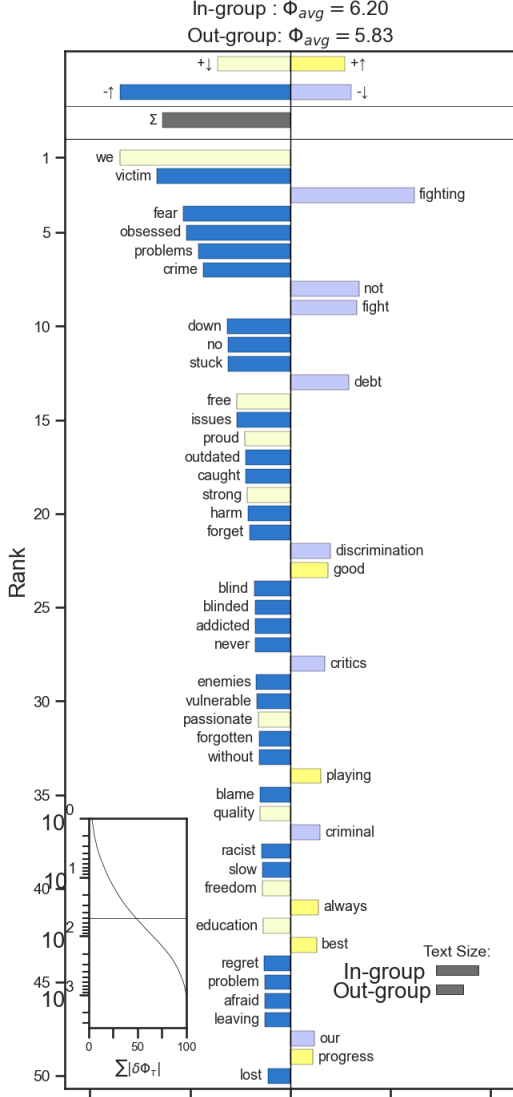


Figure 3: Word shift graph of word frequencies in happiness of ingroup and outgroup sentences. Words are ranked by their percentage contribution to the change in average happiness, Φ_{avg} . The ingroup (We are) sentences are set as the reference text T_{ref} , with the respective outgroup (They are) sentences as the comparison text T_{comp} . Individual word contributions to the shift are indicated by two symbols: $+/-$ shows the word is more/less prevalent in T_{comp} than in T_{ref} . Black and gray fonts encode the $+$ and $-$ distinctions, respectively. The left inset panel shows how the ranked 3,686 labMT 1.0 words combine (word rank r is shown on a log scale). The four bars on the top indicate the total contribution of the four types of words ($+↑$, $+↓$, $-↑$, $-↓$). Relative text size is represented by the areas of the gray squares (Gallagher et al., 2020b)

peared less frequently in outgroup sentences than in ingroup sentences, contributing positively to the sentiment balance. Overall, the decrease in average sentiment (Φ_{avg}) of 0.47 for outgroup sentences is primarily attributable to the greater presence of negative words ($-↑$) and the reduced prevalence of positive terms ($+↓$).

4 RQ2: How does ‘us versus them’ bias vary across model personas (conservative versus liberal)?

4.1 Method

To examine how partisan identity bias emerges in LLMs, we conditioned each model with explicit political personas. Specifically, models were prompted with persona instructions (e.g., “Consider yourself a Conservative/Liberal”) in the context of U.S. politics using prompt 6 and asked to generate 2,000 ingroup (“We are”) and outgroup (“They are”) sentences. Using the consistent us versus them scoring prompt 8, each sentence was evaluated for ingroup–outgroup bias. Representative examples of generated sentences with their corresponding bias scores are shown in Appendix A.6, Table 13. The distribution of bias scores for LLM-generated ingroup and outgroup sentences is presented in Appendix A.4, Figure 18.

4.1.1 Embedding Regression

We use an embedding–regression framework based on À la Carte on text (ALC) (Rodriguez et al., 2023; Khodak et al., 2018; Arora et al., 2016), which represents a focal word as a linear transform of the mean embedding of its context. This yields a transparent and fixed embedding space (affine in observed contexts), supports few-shot induction for sparse subgroup slices, and enables instance-level inference where each occurrence becomes a row in a multivariate regression (conText), with permutation on coefficient norms (Rodriguez et al., 2023). Formally, for a focal word w , the embedding can be expressed as

$$v_w = Au_w = A\mathbb{E}_c[u_{wc}], \quad (2)$$

where u_{wc} denotes the embeddings of context words, A is a learned transformation matrix, and the expectation is taken over contexts c . Because A is constant, the transformation can be integrated into a regression framework that models the embeddings conditionally on covariates such as party affiliation.

Operationally, we treat each observed instance of a focal word as a $1 \times D$ embedding vector (where D is the embedding dimension). Stacking across all occurrences yields an $n \times D$ outcome matrix Y . We then regress this outcome on a set of covariates X using the multivariate model:

$$Y_{n \times D} = X_{n \times (p+1)} \beta_{(p+1) \times D} + E_{n \times D}, \quad (3)$$

where X includes a constant and relevant covariates (e.g., binary indicators for party membership), β represents regression coefficients, and E is the residual error. In the simplest case, where X consists of a constant and a party indicator, β_0 corresponds to the average embedding of the focal word for non-Republican contexts, and $\beta_0 + \beta_1$ corresponds to the embedding in Republican contexts. This provides a model-based estimate of meaning across groups.

For each token occurrence, we collected its contextual embedding vector and coded the associated persona as categorical indicators (Conservatives, Liberal, or Default persona). We then fit a multivariate regression model where Liberal served as the reference group. The model can be written as:

$$Y = \beta_0 + \beta_1 \cdot \text{Conservatives} + \beta_2 \cdot \text{Default} + \varepsilon,$$

where Y is the embedding vector of a single occurrence of “we”, β_0 is the Liberal mean embedding, β_1 is the shift of Conservatives relative to Liberals, β_2 is the shift of No persona relative to Liberals, and ε is the residual. Predicted group means are therefore: Liberal = β_0 , Conservatives = $\beta_0 + \beta_1$, and No persona = $\beta_0 + \beta_2$. Pairwise semantic shifts were operationalized as the L2 norm of the differences between these estimated group means (e.g., $\|\beta_1\|_2$ for Conservatives vs. Liberals, $\|\beta_2\|_2$ for No persona vs. Liberals, and $\|\beta_1 - \beta_2\|_2$ for Conservatives vs. No persona).

To assess statistical reliability, we applied two complementary procedures: (i) non-parametric bootstrap resampling (1,000 replicates) to derive 95% confidence intervals around each distance, and (ii) permutation testing (1,000 label shuffles) to compute p -values for the null hypothesis of no persona effect.

4.2 Results and Discussion

Figure 4 illustrates the ingroup solidarity (μ_{ingroup}) and outgroup hostility (μ_{outgroup}) across five LLMs prompted with Liberal and Conservative personas

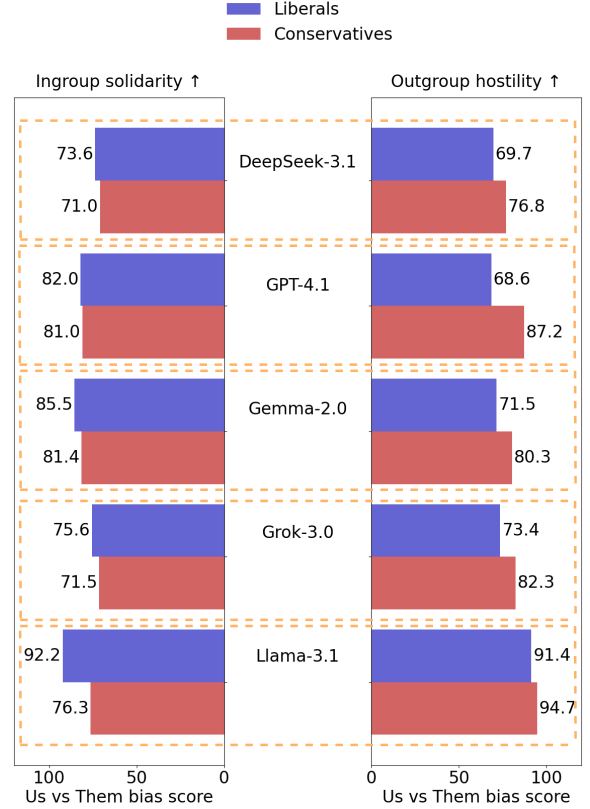


Figure 4: Ingroup solidarity (μ_{ingroup}) and outgroup hostility (μ_{outgroup}) across LLMs prompted with two partisan personas (Conservative and Liberal) in the U.S. political context.

in the U.S. political context. Across all LLMs, Conservatives consistently display higher outgroup hostility than Liberals, suggesting models have learned to associate specific identities with specific outlooks (in this case, Conservatism with a more negative outlook towards opposing groups). For instance, GPT-4.1 and Llama-3.1 show pronounced partisan divergence, with Conservative prompts producing markedly higher μ_{outgroup} hostility scores (87.2 and 94.7, respectively) than Liberal prompts (68.6 and 91.4). In contrast, DeepSeek-3.1 and Grok-3.0 show relatively balanced ingroup solidarity across personas but still exhibit higher Conservative outgroup hostility (76.8 and 82.3) compared to Liberals (69.7 and 73.4). Liberal personas, however, generally show stronger or comparable ingroup solidarity (μ_{ingroup}) across models—most notably in Llama-3.1 (92.2 vs. 76.3) and Gemma-2.0 (85.5 vs. 81.4), indicating a more affirming but less antagonistic self-representation. A logistic regression model quantifies the shift in ‘us versus them’ bias under political persona conditioning rel-